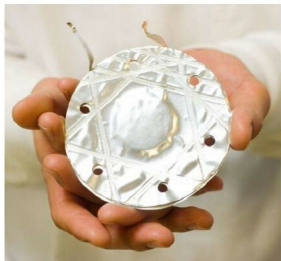


exploited with return-oriented programming.

The research team included scientists from the Universities of California, San Diego, Michigan and Princeton.

Small batteries that trap massive solar energy

Researchers at Utah-based Ceramtec have created a disk that can store up to 20 kilowatt hours of solar energy, which is enough to power a house for 24 hours.



Ceramtec solar disk can store 20 kilowatt hours of solar energy

The new battery runs on sodium-sulphur at less than 93.3°C (200°F). The secret is a thin ceramic membrane that is sandwiched between sodium and sulphur. Only positive sodium ions can pass through, leaving electrons to create electrical current.

The batteries will be ready for market testing by 2011.

Flexible, transparent LED displays

US researchers have made large-scale flexible display screens that can be stretched to fit the contours of a bus and are transparent enough for riders to see through them.

The thin, light screens can be used to make brake light indicators for cars, health monitors and imaging devices that wrap around a patient like a blanket. These combine the scale and durability of light-emitting diodes

(LEDs) used to make flat, lighted billboards, with the flexibility of screens made from organic carbon-containing materials.

The researchers built the LEDs on a thin layer of film, later dissolved the film by a chemical and then affixed tiny plastic tabs on two corners to ensure that the LEDs did not wash away in the chemical bath. A special stamping technology was used to deposit and assemble the inorganic LEDs onto glass, plastic or rubber surfaces.

Video ads in print media

Soon readers will get to see video advertisements in printed magazines. A magazine called Entertainment Weekly will be embedded with a video player that will run advertisements for CBS shows and Pepsi. The advertisements resemble greeting cards that make sounds and play music. A small screen starts playing automatically as the page is flipped open. A speaker's voice is also embedded below it.

The chip technology used is similar to the ones used in musical or singing greeting cards and activates when the page is turned. Each chip can hold up to 40 minutes of video.

Microchips based on DNA structure

International Business Machines (IBM), together with the California Institute of Technology, is developing microchips based on DNA structure. As designers struggle to cut costs, chipmakers attempt to develop smaller chips to address this challenge. Artificial DNA nano structures provide a cheaper framework to build microchips, which are used in many electronics devices like mobile phones.

Biological structures like DNA offer some reproducible, repetitive kinds of patterns that can be leveraged in semiconductor processes.

Nanotube sensor to detect nitric oxide gas

Massachusetts Institute of Technology (MIT) has developed a carbon nanotube sensor to detect nitric oxide gas. Nitric oxide is difficult to detect because it is very unstable. Monitoring nitric oxide levels in living cells will help researchers figure out its role in cancer and other diseases. It would also allow closer study of nitric-oxide-releasing cancer drugs. Biologists could use these sensors to study the effects of nitric oxide on the brain, where it acts as a neurotransmitter.

The researchers coated carbon nanotubes with a polymer designed to specifically attract nitric oxide. The polymer is wrapped tightly enough so that only small molecules can get through to bind the nanotube, and nitric oxide is strongly attracted to the nanotube by an extra pair of electrons passed from the polymer to the nanotube. The sensor is activated by near-infrared light, which easily penetrates the human body.

The team plans to start testing the sensors in living animals and is working on similar sensors that can detect molecules other than nitric oxide.

New sandwich technique for molecular electronics

A sandwich technique devised by scientists working at the National Institute of Standards and Technology (NIST) may prove useful for computer chip designers who have been working on molecule-sized electronic components.

The research team, which includes collaborators from the University of Maryland, has found a simple method of sandwiching organic molecules between silicon and metal—two materials fundamental to electronic components. By doing so, the team may have overcome one of the principal obstacles in creating switches made from individual molecules, which are perhaps the ultimate in miniaturisation for the electronics industry.

Mariona Coll Bau, a materials sci-